



## Analytics Applications — SP26 Project Proposal

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### 1. Datasets

- **MO 2020 Election Results.csv** (*Source: Harvard Dataverse*): Precinct-level vote totals for analyzing turnout and candidate performance.
  - **Key Variables**
    - STATEFP
    - COUNTYFP
    - NAME (precinct name)
    - Voting results
- **MO 2020 Polling Locations.csv** (*Source: MIT Election Data Lab*): Polling addresses to map locations and evaluate resource allocation across precincts.
  - **Key Variables**
    - county\_name
    - precinct\_id
    - precinct\_name
    - polling\_place\_id
    - name
    - address
- **MO 2020 Census Datasets** (*Income, Education, Race, Sex by Age, Commute*) (*Source: ACS 2016–2020*): A collection of county-level demographic and economic CSVs used to analyze socioeconomic factors, estimate eligible voter populations, and assess potential transportation barriers to voting.
  - **Key Variables**
    - GEO\_ID
    - (County) NAME
    - income estimates
    - education attained
    - race population
    - age and sex
    - commute time
- **MO 2020 Precincts Shapefiles** (*Source: U.S. Census TIGER/Line*): Geographic boundaries for spatial analysis and mapping of voting districts.
  - **Key Variables**
    - STATEFP20
    - COUNTYFP20
    - VTDST20
    - GEOID20
    - NAME20

### 2. Problem Statement

- Missouri has no quantified measure for the distribution of their voting resources, and analysis could solidify an objective approach to ensuring fair voter representation.



### 3. Decision Context

- This data can inform **state government officials** where they should **allocate voting resources** so that we can **reduce voter disenfranchisement**.

### 4. Analytic Task

- This is a **predictive** and **prescriptive** analysis project.

### 5. Feasibility Assessment

- This can be a **standard regression** and **optimization** problem that is **expanded as necessary**.

### 6. Expected Challenges

- Temporal and geographic **data gathering** may be challenging. **Engineering** a solution for storage, ingestion, transformation, and final service to end users will take careful consideration.